

Project DYESKIT

Inclusive Village Wellness and Sustainable Future using AI

Application Plan: Data Collection, Scoring, Insights & AI Platform

Draft v0.3 | 22 September 2026

Status: For team review and decisions

Contents

1. Purpose of this document

2. What we are building

3. Users and roles

3.1 Permission matrix (three access levels)

4. Sign-up, sign-in and access

4.1 Field researchers and village heads (mobile)

4.2 Admin, supervisors and analysts (web)

4.3 Households

4.4 Identity documents

4.5 Consent

4.6 Session security

5. How each role uses the system

5.1 Admin: setting up

5.2 Field researcher: in the village

5.3 Village head

5.4 Supervisor: reviewing

5.5 Analyst

6. Editing data and version control

6.1 Answers

6.2 Questionnaires

6.3 Scoring rules

6.4 Deleting data

7. Data storage and security

7.1 Main records to store

7.2 How data is organised for easy adding, editing and downloading

7.3 Downloading data

8. Turning data into insight

8.1 Scoring engine (fixed rules, not AI)

8.2 Dashboards

8.3 AI assistant (Claude API)

8.4 Later: village resource balance sheet

9. Languages and field realities

10. Suggested technology

11. Architecture: app, website and shared backend

11.1 Built once, shared by both (about 60% of the work)

11.2 Genuinely separate (about 40%)

11.3 The website is three things in one project

11.4 One repository (monorepo)

11.5 Why not build only a website (PWA)?

12. What to buy, and how

12.1 Buy now (pilot, first 3 months)

12.2 Buy later, not now

12.3 Field equipment

12.4 The order to do it in

12.5 Rules that save money and trouble

12.6 Cost summary

13. Phases

14. Decisions needed from the team

Appendix: issues found in the source documents

1. Purpose of this document

Project DYESKIT will collect wellbeing data from households and villages across Ladakh and turn it into scores, visualisations and AI-assisted insights. This document sets out how the application will work: who uses it, how people sign in, how data is collected, edited and stored, how it becomes insight, and the order in which we will build it.

It builds on the project proposal, the Ladakh questionnaires (v1, Regrouped, Merged ESS, v2), the One Earth scoring and threshold documents, the Village-Level AI instructions and the 788-field Data Architecture Map. Where those sources conflict or do not fit Ladakh, this plan makes a recommendation and lists the decision in Section 14.

2. What we are building

Three parts that share one backend:

Part	Who uses it	What it does
Field app (Android, works offline)	Field researchers, village heads	Collects household surveys, village data, field notes, photos and audio
Admin and analysis web portal	Project team, supervisors, analysts	Manages users, villages and questionnaires; reviews data; dashboards; reports; exports
Insight layer (scoring + AI)	Admin, analysts, later planners and villages	Calculates scores, shows charts, flags problems, answers questions in plain language

Main design choice: in phase 1, **trained field researchers collect the data**, not households on their own phones. The questionnaire is designed to be read out by an interviewer, many people in Ladakh have low digital literacy, and villages lose connectivity in winter. The field app must therefore work fully offline and sync when there is signal.

3. Users and roles

Role	Can do	Cannot do
Super Admin (core team)	Everything: users, villages, questionnaire versions, scoring rules, exports	—
Supervisor	Approve researchers, assign villages, review and approve submissions, approve corrections	Change scoring rules
Field Researcher ("Moderator" in the app spec)	Collect data only in assigned villages, edit own drafts, request corrections	See other villages, or personal details after sync

Village Head	Fill in the village module (facilities, water, land, energy, governance)	See household answers
Analyst / Researcher	Dashboards, anonymised exports, AI questions, field notes	See names, phone numbers or ID details
Viewer (government, NGOs, later villages)	Village-level dashboards only	See anything about a single household

This follows the app specification: an admin assigns one or more villages to each moderator and can reassign them later. Access is enforced on the server, not just hidden in the app.

3.1 Permission matrix (three access levels)

In simple terms there are three access levels: **Admin** (read, write, delete, manage), **Data Collector** (add data through the app) and **Viewer / User** (read-only). Supervisor, Village Head and Analyst are variations of these with narrower scope.

Permission	Admin	Data Collector (field app)	Viewer / User
Login	Web: email + password + two-factor code	Mobile: phone + OTP, after admin approval	Web: email + password, invited by admin
View data	Everything	Only records from assigned villages	Dashboards and reports, read-only
Add data	Yes	Yes, through the app	No
Edit data	Any record (change is logged)	Own drafts only, before submitting	No
Delete data	Yes (goes to recycle bin first)	No	No
Download data	Full data with filters, incl. personal details	No	Anonymised, filtered data (if admin allows)
Manage users, questions, scoring	Yes	No	No

4. Sign-up, sign-in and access

4.1 Field researchers and village heads (mobile)

- Sign up with mobile number and OTP.
- The account shows "**Review in progress**" until a supervisor approves or declines it (as in the app spec). Optional upload of a supporting document.
- After approval, the user sets a PIN or uses fingerprint so they can unlock the app offline.

4.2 Admin, supervisors and analysts (web)

- Email and password with **two-factor authentication**.
- Accounts are created by invitation from an admin only; there is no open sign-up.

4.3 Households

- Households **do not log in** in phase 1.
- Each household gets a system ID, printed as a QR code on a card, so a return visit next year links back to the same record.

4.4 Identity documents

- **Do not collect Aadhaar numbers.** The app spec asks for Aadhaar verification; we recommend dropping it. It is legally restricted, a privacy risk, and not needed for wellbeing research. Record village, household ID and GPS location instead.

4.5 Consent

- A consent screen opens before every interview, using the consent script from questionnaire v2.
- The app records yes/no, time, researcher and questionnaire version. If the answer is "no", the survey does not open.
- Sensitive items (e.g. substance use) can be handed to the respondent to answer privately.

4.6 Session security

- Sessions time out after inactivity.
- Admins can deactivate a user and wipe that account's data from a lost phone at its next sync.

5. How each role uses the system

5.1 Admin: setting up

1. Create districts (Leh, Kargil), blocks and villages, with GPS location and the local land unit.
2. Publish the questionnaire version (e.g. v2.0) with its translations.
3. Approve researchers and assign villages.
4. Set the sample target per village (at least 30% of households or 10, whichever is more).
5. Watch progress: surveys done per village, sync status, data-quality alerts.

5.2 Field researcher: in the village

1. Download assigned villages and the questionnaire while online.
2. Offline: register the household, record consent, then run the survey.

3. The app skips questions that do not apply, checks answers make sense (e.g. height 100–220 cm), and allows "don't know" or "prefer not to answer" where the questionnaire does.
4. Measure height and weight with the kit (v2 requires measured, not self-reported, values) or mark "not collected".
5. Add field notes, photos or audio, tagged by wellbeing area.
6. Save as draft, then Submit. Records are stored encrypted on the phone and upload automatically when there is signal.

5.3 Village head

- Completes the village module once a year: population, land, water sources, energy, roads, facilities (bank, clinic, school, etc.), governance and culture.
- Fixed facts (facility counts, budgets) should later be pre-filled from official records and only confirmed by the village head.

5.4 Supervisor: reviewing

- Checks each submission and approves it or sends it back with a comment.
- The system auto-flags problems:
 - interviews completed too fast
 - identical answers throughout ("straight-lining")
 - GPS location outside the village
 - missing sections
- Runs a 10% re-interview check to measure data reliability.

5.5 Analyst

- Uses dashboards, compares villages, reads field notes and asks the AI assistant questions.
- Exports anonymised data as CSV or SPSS (the questionnaires already include SPSS coding).

6. Editing data and version control

6.1 Answers

- Researchers can edit drafts freely.
- After submission, a change needs a correction request that a supervisor approves.
- **Nothing is overwritten.** Every change is kept as a new version with who, when, old value, new value and reason.

6.2 Questionnaires

- Questions are stored in the database as versioned settings, not built into the app's code.

- An admin can change wording, options or translations and publish a new version without releasing a new app.
- Every answer stays linked to the questionnaire version it came from, so year-on-year comparison stays valid.

6.3 Scoring rules

- Scoring rules are versioned the same way.
- If a rule changes (e.g. a new BMI cut-off), scores are recalculated from the raw answers and old results are kept.
- Raw answers are never replaced by scores.

6.4 Deleting data

- Only admins can delete.
- Deleted records go to a **recycle bin for 30 days** and can be restored; after that they are permanently removed.
- Permanent deletion is also used when a household withdraws consent (their right under the DPDP Act).
- Every delete and restore is recorded in the audit log.

7. Data storage and security

Item	Plan
Main database	PostgreSQL with PostGIS (for maps and GPS)
Files	Photos, audio and uploaded documents in separate secure file storage (S3-compatible)
Hosting	India data-centre region; suits India's Digital Personal Data Protection (DPDP) Act and eases government partnerships
Personal details	Names, phone numbers and GPS kept in a separate, locked table. Analysts and the AI see only anonymised household IDs
Encryption	On the phone, in transit (HTTPS) and on the server
Backups	Daily automated backups with tested restore
Audit	Every login, edit, approval and export is logged

7.1 Main records to store

Area	Records
Places	District, block, village (with GPS boundary and local land unit)
People	Household, household member, respondent

Access	User, role, village assignment, approval status
Collection	Consent record, questionnaire version, question, submission, answer
Qualitative	Field note, interview transcript, photo/audio file, dimension tag
Scoring	Scoring version, item score, dimension score, household score, village score
Governance	Correction request, review comment, audit log, export log

7.2 How data is organised for easy adding, editing and downloading

- **Every answer is stored as its own row**, tagged with household, village, question code and questionnaire version. This makes it easy to add records from the app, edit a single answer, or add new questions later without redesigning the database.
- **A ready-made table with one row per household** (one column per question, using the SPSS variable names from the questionnaire) is refreshed automatically. This is what gets downloaded, so files open directly in Excel or SPSS.
- Scores (per area, per household, per village) are kept in their own tables and can be downloaded alongside the answers.
- Nothing is edited silently: every change keeps the old value, who changed it and when.

7.3 Downloading data

Option	Detail
Scope	Whole dataset, or filtered
Filters	District, block, village, date range, questionnaire version, data collector, submission status, gender, age group, religion, wellbeing area, score band
What to include	Raw answers, scores, field notes, or all of them
Formats	CSV, Excel (.xlsx), SPSS (.sav), JSON
Codebook	Every download includes a codebook explaining each column and its codes
Privacy	Personal details are removed unless an admin downloads the data
Large files	Prepared in the background; the user gets a download link when ready
Audit	Every download is logged (who, when, which filters)

8. Turning data into insight

8.1 Scoring engine (fixed rules, not AI)

- Answer scores → area (dimension) scores → household wellbeing score → **Village Wellbeing Index (VWBI)**.

- We recommend the **0–1 graduated scoring from questionnaire v2**: it keeps more detail than the 1 / 0.5 / 0 scale and already uses Indian thresholds (ICMR BMI cut-offs, IPHS hill-area norms, BIS drinking-water standard, ₹ income bands).
- Results are shown as 0–1 and on the familiar 1–7 band (Foundational Support Needed → Thriving Well-Being).
- Villages below the minimum sample show "insufficient data" instead of a score.

8.2 Dashboards

- Map of Ladakh coloured by village score
- Village scorecard with a 7-area radar chart
- Village compared with the district and survey-wide average
- Breakdowns by gender, age group, religion and Leh/Kargil
- Flag table for any area below "Basic" (under 43%)
- Villagers' top development priorities
- Survey coverage and data-quality panel
- From year 2: trends over time

8.3 AI assistant (Claude API)

- Answers plain-language questions, e.g. "Which villages in Kargil show water stress together with low emotional wellbeing?"
- Works from scores and anonymised field notes and **shows the numbers and notes behind every answer**.
- Drafts strengths/weaknesses (SWOT) analyses and village profiles for the Wellness Atlas.
- A human reviews AI output before anything is published, as the proposal requires. The AI never sees personal details.

8.4 Later: village resource balance sheet

- Water, energy, food, housing, land and other resources tracked as assets, needs and flows, as described in the One Earth documents.
- Thresholds must be adapted to Ladakh first (e.g. the 60% forest-cover target does not fit a cold desert).

9. Languages and field realities

- Screens in English and Hindi.
- Questionnaire also in Ladakhi (Bhoti script) and Urdu for Kargil.
- Audio playback of questions for respondents who cannot read.
- Built for older, low-cost Android phones, cold weather and battery drain.
- Researchers must be able to work for weeks without signal.

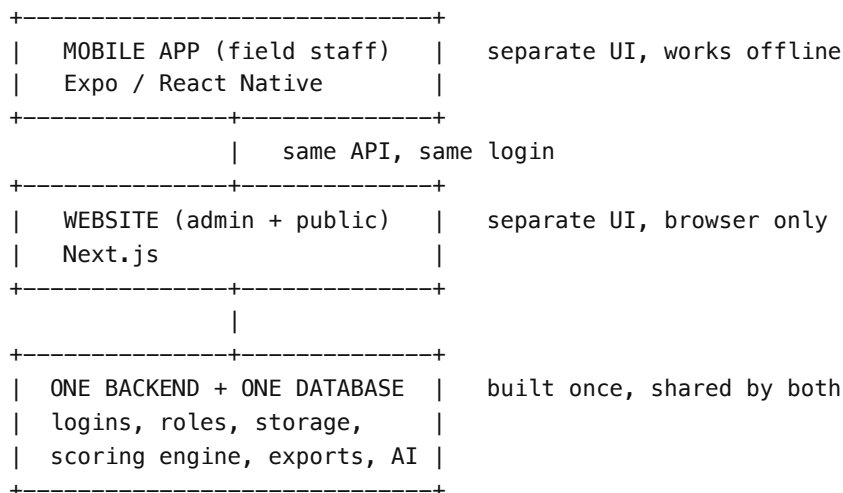
10. Suggested technology

Layer	Choice
Field app	React Native (Expo) with an encrypted offline database (SQLite) and a sync queue
Web portal	Next.js
Backend	Python (FastAPI) — scoring, statistics and AI work are easiest in Python
Database and files	PostgreSQL + PostGIS; S3-compatible file storage
Charts and maps	A standard chart library plus Leaflet for maps
AI	Claude API

Fallback if the 4-month timeline is tight: collect the pilot data with KoboToolbox (free, works offline) while we build the dashboard and scoring, then move to our own app.

11. Architecture: app, website and shared backend

The mobile app and the website are **not two separate programs**. They are two user interfaces on one backend, built in one repository.



11.1 Built once, shared by both (about 60% of the work)

- Database and all the data
- Login and permissions (admin / data collector / viewer)
- The API that both front-ends call
- Scoring engine, exports, AI assistant
- A shared code package: question definitions, scoring rules, validation and shared types

11.2 Genuinely separate (about 40%)

- **Mobile app:** survey screens, offline storage, sync, camera, GPS, audio. This is the hard part, and it must be an app because it has to work for weeks with no signal.
- **Website:** admin screens, dashboards, charts, maps, review and export screens. These work far better in a browser on a large screen.

The screens cannot be shared, because the two jobs are different. But the rules, scoring and data handling are written only once.

11.3 The website is three things in one project

Part	Who sees it	Login needed
Public pages (about the project, contact, privacy policy)	Anyone	No
Admin and analysis portal	Project team	Yes
Village / government dashboards	Invited viewers	Yes

11.4 One repository (monorepo)

dyeskit/

apps/mobile	Expo app
apps/web	Next.js website + admin portal
packages/shared	questions, scoring, types (used by both)
supabase/	database structure

One repository, one deployment pipeline, one set of rules, and no copy-pasted logic drifting apart. Total effort is roughly 1.4 times a single product, not 2 times.

11.5 Why not build only a website (PWA)?

A single website that also works offline on phones is cheaper and skips the Play Store. But phone browser storage is less reliable for weeks of disconnected work, background sync is weaker, and data can be lost if the browser clears storage. Survey data cannot be re-collected, so the pilot keeps a separate app whose whole job is to not lose data.

12. What to buy, and how

Software and hosting for the pilot cost roughly ₹5,000–8,000 a month. **People are the real cost.** Prices below are approximate (₹88 = \$1) and change; check before purchase.

12.1 Buy now (pilot, first 3 months)

Item	Where	Cost
Domain name (e.g. dyeskit.org / .in)	Cloudflare Registrar or Namecheap	₹1,000–1,500 / year

Backend + database + storage + login: Supabase Pro, Mumbai region	supabase.com	\$25/mo ≈ ₹2,200
Website and admin portal hosting: Vercel	vercel.com	Free to start, \$20/mo later
Claude API (AI assistant)	Anthropic Console, pay as you go	≈ ₹2,000–4,000/mo at pilot volume
Project email: Google Workspace or Zoho Mail	google.com/workspace, zoho.com	₹75–160 per user/month
Password manager: Bitwarden	bitwarden.com	Free, or ₹850/year
Code hosting: GitHub	github.com	Free (private repositories)
Error alerts: Sentry; uptime alerts: UptimeRobot	sentry.io, uptimerobot.com	Free tiers
Maps: OpenStreetMap + Leaflet; SSL: Cloudflare	—	Free

Pilot total: about ₹5,000–8,000 a month, plus ₹2,000 one-time. At full rollout (113 Leh villages) expect ₹15,000–30,000 a month, mostly larger database, media storage and AI use.

12.2 Buy later, not now

Item	Cost	When
Google Play developer account	\$25 one-time	Only for public release. For own field staff, install the app file directly or use Firebase App Distribution (free).
Apple developer account	\$99 / year	Only if iPhones are ever needed. Skip.
Expo EAS build plan	\$19 / month	Only if the free build quota runs out.
AWS or Azure	₹15,000+ / month	Only if a government partner requires their empanelled cloud.
SMS OTP service	See warning below	Only if many users need phone OTP.

Warning about SMS in India: sending OTP messages requires DLT registration with the telecom regulator, which takes weeks, costs a few thousand rupees and adds per-message charges. For a team of 10–20 field staff this is not worth it — have the admin create their accounts with a password instead, and add phone OTP later if households ever log in themselves.

If publishing to the Play Store later, open an organisation account, not a personal one: personal accounts must run a closed test with a group of testers for 14 days before publishing. Organisation accounts need a DUNS number for the institution, which is free but takes a couple of weeks — start it early.

12.3 Field equipment

Questionnaire v2 requires measured height and weight, so this matters.

Item	Cost each	How many
Android phone, 4 GB RAM or more	₹12,000–15,000	1 per field researcher, plus 1 spare
Digital weighing scale	₹1,500–3,000	1 per team
Height measure (stadiometer or tape)	₹1,000–2,500	1 per team
Power bank, 20,000 mAh	₹2,000	1 per researcher
Waterproof bag, spare cables	₹2,000	1 per researcher

For 2 field researchers: about **₹40,000–50,000 one-time**.

12.4 The order to do it in

1. **Create a project email account first** (e.g. tech@dyeskit.org) and register everything else with it, never a personal address. Store every password in Bitwarden, shared with two people.
2. Buy the domain and point it at Cloudflare for free DNS and SSL.
3. Create the GitHub organisation and a private repository.
4. Create the Supabase project in the Mumbai region — one account gives database, login, file storage and role permissions.
5. Hire or assign the developer; hand over this plan and the final questionnaire.
6. Build in this order: login and roles → village and user setup → offline survey → sync → supervisor review → scoring → dashboard → export.
7. Write the privacy notice and consent text while development runs.
8. Distribute the app to field staff through Firebase App Distribution, skipping the Play Store for the pilot.
9. Pilot in 2–3 villages, then fix what breaks.
10. Only then open a Claude API account and add the AI assistant, once there is real data for it to work with.

12.5 Rules that save money and trouble

- Set a billing limit and alerts on every paid account. Cloud bills grow quietly.
- Take your own nightly database backups (about ₹500/month), not only the provider's.
- Keep data in the India region on every service.
- Own your data: everything above stores data in standard PostgreSQL, which can be moved elsewhere.
- Pay monthly at first; switch to yearly (usually 15–20% cheaper) once a service proves itself.
- Do not buy a design tool, CRM or analytics suite. They are not needed.

12.6 Cost summary

	One-time	Monthly
Software and hosting	≈ ₹2,000	₹5,000–8,000 (pilot); ₹15,000–30,000 (full rollout)
Field equipment	₹40,000–50,000	—
Development (the real cost)	₹3–6 lakh for phase 1	—

The proposal budgets ₹4 lakh for technical engineers and ₹4 lakh for tech and software. That supports one good full-stack developer for about 4 months (₹60,000–1,00,000 per month) plus a part-time mobile developer, with room left for hosting.

13. Phases

Phase	Time (rough)	Deliverables
0. Decide and design	2 weeks	Final questionnaire and scoring model, translations, screen mockups, data structure
1. First working version	6–8 weeks	Sign-in and roles, villages and assignments, offline survey with consent, sync, supervisor review, scoring engine, basic dashboard, CSV/SPSS export
Pilot	2 weeks	10–15 households in 2–3 villages (different altitudes, Leh and Kargil); fix problems; reliability checks (Cronbach's α per area)
2. Full rollout	Next ~2 months	Village-head module, field notes/audio, maps, village comparison, AI assistant, draft Atlas profiles
3. Scale up	Later	Resource balance sheet, household self-service app, viewers for villages and government, links to government data and satellite maps, yearly trends, 113 → 256 villages

14. Decisions needed from the team

1. **Questionnaire:** v2, or the team-edited v1/Regrouped version? (Recommended: v2 plus the team's useful edits.)
2. **Scoring:** 0–1 graduated (recommended) or the proposal's 28-indicator 1 / 0.5 / 0 model?
3. **Collection:** field researchers only in phase 1 (recommended), or households on their own phones too?
4. **Hosting and budget:** own cloud account (AWS or Azure, India region) or a managed service?
5. **Languages** needed on day one.
6. **Scope:** Ladakh only, or keep the platform ready for multiple countries (India, Bhutan, Malaysia) as the original app spec intends?

Appendix: issues found in the source documents

- Two scoring models disagree (28-indicator 1 / 0.5 / 0 vs 0–1 graduated).
- The proposal's formula " $VWBI = WBI \div \text{number of people surveyed}$ " is wrong; it should be an average.
- Team edits that need fixing in v1/Regrouped:
 - C6 reads "I often visit outside Ladakh if yes (reason text)" but is still a 1–5 rating
 - D5 is cut off: "...youth are actively engaged"
 - Religion options reduced to Buddhist / Muslim / Other
- Platform thresholds are Malaysian or Bhutanese (60% forest cover, ringgit income bands, Malaysian agencies).
- App dropdowns still include Dzongkha, Malay, tropical diseases and fishing.
- The app has no spiritual questions; about 30 wellbeing questions and the community-infrastructure section are missing.
- Naming is inconsistent: "Financial" vs "Occupational"; the proposal sometimes says "vocational".
- Open note in the Data Architecture Map: add "seeds available in the region" (recommended by John and Sneha).